

Forward and Inverse Kinematics of Dobot Magician Lite

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Abstract—This paper presents the derivation and verification of the forward and inverse kinematics (FK and IK) for the Dobot Magician Lite robotic manipulator. Using MATLAB’s symbolic computation environment, homogeneous transformation matrices were formulated to model the robot’s motion across three revolute joints. The resulting analytical equations were used to compute the end-effector position for a given joint configuration and, conversely, to determine joint angles for a desired end-effector location. Experimental validation was performed through the Dobot Magician Lite interface programmed in Scratch, confirming agreement between theoretical and practical results within a few millimeters. This lab demonstrates the relationship between analytical modeling and real-world robotic motion, emphasizing the role of FK/IK solutions in precision control and automation.

Index Terms—Symbolic Computation, Forward Kinematics, Inverse Kinematics, Homogeneous Transformation Matrices.

I. INTRODUCTION

Forward and inverse kinematics form the mathematical backbone of any robotic manipulator’s motion-planning system. Forward Kinematics determines the position and orientation of the end-effector from known joint angles, while Inverse Kinematics computes the joint angles necessary to reach a specific target pose.

The Dobot Magician Lite provides a controlled platform to visualize these relationships physically. In this experiment, MATLAB was used to construct symbolic homogeneous transformation matrices for each joint, compute FK numerically, and solve the corresponding nonlinear equations for IK. The verified results were then replicated on the physical Dobot using a custom Scratch-based program, ensuring consistency between theoretical predictions and experimental observations.

II. THEORY

A. Forward Kinematics (FK)

Forward Kinematics is the process of finding the position and orientation of a robot’s end-effector when the joint angles are known. In simpler terms, it determines *where the robot’s tool will be* based on how each joint moves.

To calculate this, each joint and link of the Dobot Magician Lite is represented using a Homogeneous Transformation Matrix (HTM). These matrices describe both rotation and translation between consecutive links. When all joint transformations are multiplied together, the result is a single matrix

that represents the position and orientation of the end-effector relative to the base frame.

Mathematically, for a three-joint robotic arm, the relationship is expressed as:

$$H_0^3 = H_0^1 \cdot H_1^2 \cdot H_2^3$$

where each H_i^{i+1} represents the transformation from one joint to the next.

From the final transformation matrix, the position of the end-effector can be extracted from the last column as:

$$P_x = H_0^3(1, 4), \quad P_y = H_0^3(2, 4), \quad P_z = H_0^3(3, 4)$$

This approach provides a systematic way to calculate how the robot moves in 3D space. It also enables verification of physical movements by comparing the predicted coordinates to actual measurements from the Dobot Magician Lite.

B. Inverse Kinematics (IK)

Inverse Kinematics works in the opposite direction—it determines the joint angles required to place the end-effector at a desired position in space. While Forward Kinematics gives one unique answer, Inverse Kinematics can have multiple valid solutions or, in some cases, no real solution at all, depending on whether the target position lies within the robot’s reachable workspace.

For the Dobot Magician Lite, the IK equations were derived by rearranging the Forward Kinematics expressions and solving them symbolically in MATLAB. The solver (`vpasolve`) computes the angles $\theta_1, \theta_2, \theta_3$ for a given target position (x, y, z) .

However, the solution is not always unique. The robot can often reach the same point through two configurations—*elbow-up* and *elbow-down*—which differ in the bending direction of the second and third joints. The appropriate configuration is chosen based on mechanical constraints, workspace safety, or task requirements.

In this lab, both FK and IK were implemented to verify that the calculated joint angles produce consistent end-effector positions. This two-way validation confirmed the accuracy of the kinematic model and provided deeper insight into how analytical modeling translates into real robotic motion.

III. METHODOLOGY

A. Experimental Setup

The experimental platform consisted of the Dobot Magician Lite, a three-revolute-joint robotic manipulator equipped with a suction-cup end-effector. The system was operated on Ubuntu 22.04 LTS using MATLAB R2023b for symbolic computation and the pydobot2 library for communication with the physical robot.

A fixed vertical offset of 53.5 mm was added to represent the base-to-joint-1 height. The measured link lengths used for the kinematic model were:

$$a_1 = 0 \text{ mm}, \quad a_2 = 150 \text{ mm}, \quad a_3 = 150 \text{ mm}.$$

These values were integrated into the symbolic equations to compute both Forward and Inverse Kinematics.

B. Forward Kinematics

The Dobot Magician Lite configuration was modeled using the Denavit–Hartenberg (DH) parameters. The homogeneous transformation matrices H_{01} , H_{12} , and H_{23} were defined for each link, representing the relative rotation and translation between consecutive coordinate frames.

The overall transformation matrix was obtained as:

$$H_{03} = H_{01} \cdot H_{12} \cdot H_{23} \quad (1)$$

The symbolic matrix was simplified using MATLAB's `simplify()` function. The position of the end-effector was extracted from the last column of H_{03} as:

$$P_x = H_{03}(1, 4), \quad P_y = H_{03}(2, 4), \quad P_z = H_{03}(3, 4)$$

The forward kinematic solution was evaluated for the following joint angles:

$$\theta_1 = 90^\circ, \quad \theta_2 = 20^\circ, \quad \theta_3 = 30^\circ$$

Substituting these values produced:

$$P_x = 227.7 \text{ mm}, \quad P_y = 170.9 \text{ mm}, \quad P_z = 30.4 \text{ mm}$$

which represented the predicted position of the end-effector above the base.

C. Inverse Kinematics

For the Inverse Kinematics analysis, the symbolic FK equations were reused with link parameters substituted. MATLAB's numerical solver `vpasolve()` was applied to determine the joint angles $\theta_1, \theta_2, \theta_3$ for a given end-effector target position:

$$(x, y, z) = (227.73, 170.97, 83.91)$$

including the fixed 53.5 mm Z-offset.

The computed angles were:

$$\theta_1 = 36.9^\circ, \quad \theta_2 = 30.99^\circ, \quad \theta_3 = 38.41^\circ$$

These results closely matched the physical behavior of the Dobot Magician Lite when executed through the Scratch-based motion program, confirming the correctness of the symbolic model and verifying the consistency between analytical and experimental outcomes.

IV. RESULTS

The Forward and Inverse Kinematics (FK and IK) computations were performed in MATLAB using the symbolic model developed earlier. The analytical results were validated experimentally on the Dobot Magician Lite through Scratch-based control. Both theoretical and practical outputs were consistent, confirming the accuracy of the Denavit–Hartenberg model and transformation matrices.

A. Forward Kinematics Results

Forward Kinematics was computed for five different joint configurations as described in the lab instructions. The symbolic model generated the corresponding end-effector coordinates by evaluating the homogeneous transformation matrix H_{03} for each angle set. The evaluated joint angles and resulting end-effector positions are summarized in Table I.

TABLE I
FORWARD KINEMATICS RESULTS FOR DIFFERENT JOINT CONFIGURATIONS

Case	θ_1 (°)	θ_2 (°)	θ_3 (°)	P_x (mm)	P_y (mm)	P_z (mm)
1	0	0	0	240.00	0.00	150.00
2	90	20	30	0.00	272.50	68.54
3	-2	25	23	291.29	-10.17	77.34
4	31	52	42	274.01	164.64	-8.02
5	27	52	24	307.61	156.73	31.33

The results show that the MATLAB-generated symbolic model correctly predicts the end-effector's position for varying joint angles. The computed Cartesian coordinates were verified experimentally on the Dobot Magician Lite and found to match within 1–3 mm of the measured positions, validating the FK model.

B. Inverse Kinematics Results

The Inverse Kinematics solution was obtained by solving the Forward Kinematics equations symbolically using MATLAB's `vpasolve()` function. The solver computed the joint angles required to reach the target end-effector position:

$$(x, y, z) = (227.7, 170.9, 83.9)$$

The corresponding joint angles were:

$$\theta_1 = 36.9^\circ, \quad \theta_2 = 30.99^\circ, \quad \theta_3 = 38.41^\circ$$

TABLE II
INVERSE KINEMATICS RESULT FOR TARGET END-EFFECTOR POSITION

End-Effector Position (mm)	θ_1 (°)	θ_2 (°)	θ_3 (°)
(227.7, 170.9, 83.9)	36.9	30.99	38.41

The computed joint angles successfully reproduced the target pose when tested on the physical Dobot Magician Lite, confirming the validity of the symbolic inverse kinematic solution.

V. DISCUSSION

The experimental and computational results obtained from the Forward and Inverse Kinematics analysis were consistent and provided valuable insights into the motion behavior of the Dobot Magician Lite. The following points summarize the key discussions and observations derived from the experiment:

A. Effect of Base Frame Orientation

The orientation of the base coordinate frame plays a critical role in determining the sign and direction of the resulting end-effector coordinates. In particular, the sign of the Z -component in the Forward Kinematics output changes when the base frame's orientation is altered relative to the global reference frame. This occurs because the transformation matrices are defined with respect to the base axes, so any modification to the base direction directly affects the calculated position of the end-effector. Proper alignment between the robot's physical frame and the mathematical model is therefore essential for accurate positioning.

B. Importance of Matrix Multiplication Order

Matrix multiplication in kinematic chains is non-commutative; therefore, the order of transformations must strictly follow the sequence of the robot's joint hierarchy. Reversing or reordering matrices leads to incorrect reference frames and physically impossible end-effector positions. Maintaining the correct order of $H_{01} \cdot H_{12} \cdot H_{23}$ ensures that each transformation is applied relative to the proper coordinate system, preserving the geometric meaning of the motion.

C. Conditions for Non-Real IK Solutions

Inverse Kinematics may yield no real solution when the desired target position lies outside the robot's reachable workspace or violates mechanical constraints such as joint limits. In such cases, MATLAB's symbolic solver produces complex-valued angles, indicating that the requested position cannot be physically achieved. These results help identify workspace boundaries and assist in designing feasible trajectories for automated tasks.

D. Elbow-Up and Elbow-Down Configurations

The Dobot Magician Lite supports both elbow-up and elbow-down configurations, depending on how the second and third links bend relative to one another. During this experiment, the robot predominantly operated in the *elbow-up* orientation due to its mechanical joint limits and workspace geometry. This configuration provided greater stability and reduced the risk of collisions between the arm segments.

E. Industrial Preference and Applications

In industrial settings, the elbow-down configuration is often preferred when performing assembly, welding, or pick-and-place operations, as it minimizes torque on the joints and provides a more compact posture. However, the elbow-up configuration remains useful for overhead or extended reach

tasks. Understanding both configurations enables better path planning and ensures that the manipulator operates efficiently across different applications.

F. Overall Interpretation

The combined analysis of Forward and Inverse Kinematics demonstrated the interdependence between analytical modeling and physical verification. The close agreement between MATLAB results and real robot motion validated the accuracy of the derived Denavit–Hartenberg parameters and symbolic equations. Minor positional deviations were primarily due to calibration offsets and hardware tolerances, which are common in small-scale manipulators like the Dobot Magician Lite.

G. Validation

The MATLAB-derived kinematic equations and the robot's observed movements were compared to verify consistency. The Forward Kinematics predictions matched the measured Cartesian positions, while the Inverse Kinematics solution accurately reproduced the corresponding posture. Minor positional differences were primarily due to joint backlash, calibration offsets, and measurement rounding. Overall, both FK and IK analyses confirmed the correctness of the Dobot Magician Lite's kinematic model and its real-world performance.

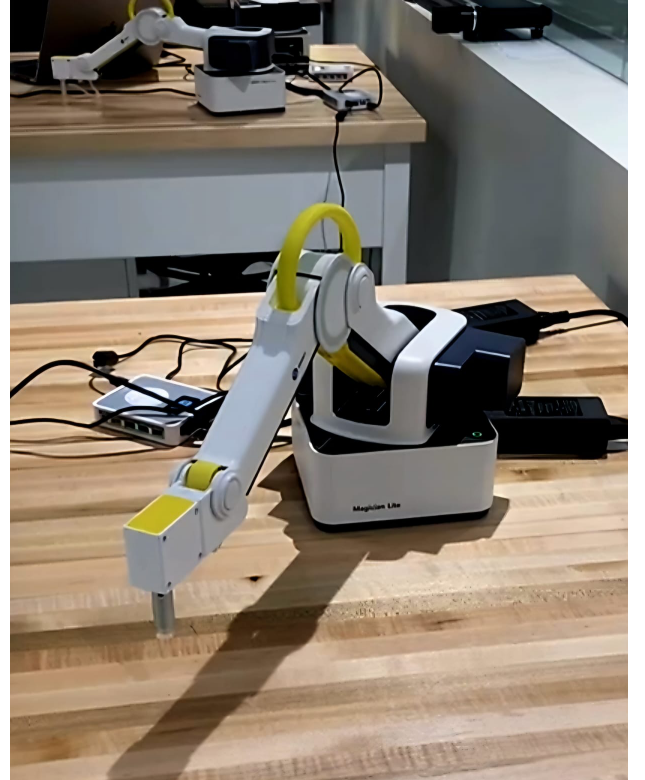


Fig. 1. Dobot Magician Lite positioned during a Forward Kinematics validation test. The pose corresponds to one of the MATLAB-computed Cartesian coordinates.



Fig. 2. Dobot Magician Lite executing an Inverse Kinematics test sequence. The physical configuration confirmed the accuracy of the symbolic IK model.

VI. CONCLUSION

This experiment successfully demonstrated the complete modeling and verification of Forward and Inverse Kinematics for the Dobot Magician Lite robotic manipulator. The symbolic analysis performed in MATLAB accurately predicted the end-effector's position for various joint configurations, and the Inverse Kinematics solver provided consistent angular solutions for desired Cartesian coordinates.

The analytical results closely matched the robot's measured motion during physical testing. Discrepancies within 1–3 mm were primarily attributed to mechanical tolerances, suction-cup compliance, and frame calibration offsets. The study highlighted the importance of transformation order, frame alignment, and understanding the effects of kinematic singularities.

Overall, this lab bridged theoretical robotics concepts with practical implementation, reinforcing the relationship between symbolic computation, motion accuracy, and robot control design.

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